# Hourly Demand Forecasting — Uber Eats Operational & Customer Intelligence System

## 1. Executive Summary

This report covers Day 5 of the 7-day Uber Eats Operational & Customer Intelligence System capstone project. The primary goal of this phase is to model and forecast hourly order volume to enable proactive fleet allocation, driver incentive planning, and delivery fulfillment management.

Using synthetic project data spanning six months (April 1 to September 30, 2026; 4,392 hourly slots), we evaluated three modeling approaches on a chronological train-test split: an **ARIMA(1,0,1)** statistical time-series model, **Meta's Prophet** forecasting engine, and a **168-hour Seasonal Naive baseline**.

Key findings from our empirical comparison include:

- **Stationarity Confirmed:** The Augmented Dickey-Fuller (ADF) test yielded a statistic of **-34.3815** , confirming that the hourly aggregated time series is stationary and suitable for linear time-series modeling without additional differencing.
- **Model Performance: ARIMA(1,0,1)** marginally achieved the best performance with a Mean Absolute Error (**MAE**) of **0.860 orders/hour** and a Root Mean Squared Error (**RMSE**) of **1.122**, closely matched by **Prophet** (**MAE: 0.866**, **RMSE: 1.124**).
- **Baseline Superiority:** Both machine learning and statistical models substantially outperformed the **Seasonal Naive baseline** (**MAE: 1.231**, **RMSE: 1.614**), achieving an error reduction of over **30%**.
- **Operational Readiness & Caveats:** While overall point prediction errors remain under 1 order/hour, the high relative percentage error (**MAPE ~31.7%**) stems from low average baseline demand (1.14 orders/hour). Furthermore, statistical non-significance of ARIMA parameters ($p > 0.05$) indicates that further parameter grid search and count-data modeling (e.g., Poisson regression) are required prior to production deployment.

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## 2. Business Context and Objective

In a two-sided food delivery marketplace like Uber Eats, anticipating hourly order demand across geographic zones is essential for maintaining operational balance:

- **Courier Supply Allocation:** Underestimating hourly demand leads to rider shortages, longer pickup delays, and order cancellations. Overestimating demand leads to low driver earnings due to courier over-saturation.
- **Kitchen Operational Flow:** Giving partner restaurants advance notice of expected order surges allows kitchens to optimize prep staff and prevent order bottlenecks.
- **Dynamic Pricing & Incentives:** Accurate short-term forecasting allows automated dispatch systems to trigger surge pricing or driver quest incentives proactively rather than reactively.

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## 3. Dataset Overview and Cleaning

The analysis was performed on the aggregated project order repository (orders.csv).

### Dataset Metadata & Integrity Verification

- **Timestamp Column:** order_timestamp
- **Analysis Period:** April 1, 2026, 00:00 to September 30, 2026, 23:00 (183 full calendar days)
- **Expected Hourly Slots:** $183 \text{ days} \times 24 \text{ hours/day} = 4,392 \text{ hours}$
- **Observed Hourly Observations:** 4,392
- **Missing Hourly Intervals:** 0 (100% complete time index)

### Cleaning & Profiling Summary

Raw individual transaction timestamps were parsed into standard ISO-8601 datetime format and validated for chronological continuity. Missing value checks confirmed zero gaps in the temporal sequence, ensuring that zero-order hours were explicitly preserved rather than omitted.

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## 4. Hourly Aggregation and Exploratory Data Analysis

Individual order logs were resampled into an hourly frequency series $Y_t$, representing total order volume per hour.

| Verified Descriptive Statistic | Metric Value |
|---|---|
| **Total Hourly Observations ($N$)** | 4,392 hours |

| Verified Descriptive Statistic | Metric Value |
|---|---|
| **Mean Hourly Demand ($\mu$)** | 1.14 orders/hour |
| **Maximum Hourly Demand** | 7.00 orders/hour |
| **Minimum Hourly Demand** | 0.00 orders/hour |
| **Missing Hours** | 0 |

### Analytical Observations

1. **Low Demand Density:** The hourly series operates at a relatively low scale ($\mu = 1.14$). A significant portion of hourly bins contain 0, 1, or 2 orders.
2. **Impact on Error Evaluation:** In low-volume count series, small absolute deviations (e.g., predicting 2 orders when actual is 1) result in high percentage errors ($100%$).
3. **Observations Requiring Visual Plot Confirmation:** While summary stats confirm a discrete range from 0 to 7 orders/hour, visual inspection of time-series plots is still required to confirm specific daily peak hours (e.g., lunch vs. dinner spikes) or day-of-week patterns (e.g., weekend vs. weekday variations).

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## 5. Time-Series Decomposition

To understand underlying patterns, time-series decomposition splits the hourly order series $Y_t$ into three constituent components:

### Key Concepts for Non-Technical Stakeholders

- **Trend ($T_t$):** The long-term direction of order volume over the 6-month period (e.g., gradual growth in platform adoption).
- **Seasonality ($S_t$):** Recurring, predictable patterns occurring at fixed intervals:
    - *Daily Seasonality (24-hour cycle):* Peak demand during lunch/dinner vs. trough demand during overnight hours.
    - *Weekly Seasonality (168-hour cycle):* Variations between weekday meal times and weekend demand spikes.
- **Residual / Noise ($e_t$):** Irregular, random fluctuations that cannot be explained by trend or repetitive seasonal cycles (e.g., random short-term demand spikes).

*Note: While classical additive decomposition models seasonal structure, exact visual confirmation of trend stability and seasonal amplitudes must be verified directly against generated decomposition plots in the project notebook.*

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## 6. ADF Stationarity Test

Stationarity is a core assumption of linear time-series models such as ARIMA. A time series is **stationary** if its mean, variance, and autocovariance remain constant over time.

### The Augmented Dickey-Fuller (ADF) Test Explained

The ADF test checks for the presence of a **unit root** (which indicates non-stationarity):

- **Null Hypothesis ($H_0$):** The series has a unit root (it is non-stationary and exhibits time-dependent structure).
- **Alternative Hypothesis ($H_1$):** The series does not have a unit root (it is stationary).

### Verified Test Results

| Test Metric | Value |
|---|---|
| **ADF Statistic** | **-34.3815** |
| **Displayed p-value** | **0.0** |
| **1% Critical Value** | -3.4318 |
| **5% Critical Value** | -2.8622 |
| **10% Critical Value** | -2.5671 |

### Statistical Interpretation

1. **Rejection of Null Hypothesis:** The test statistic (**-34.3815**) is dramatically lower (more negative) than even the **1% critical value** (**-3.4318**).
2. **Understanding the $p$-value:** The displayed $p$-value of 0.0 is due to standard numerical rounding in statistical software (statsmodels) for numbers smaller than $10^{-15}$. It should **not** be interpreted as a literal zero probability, but rather as strong empirical evidence against $H_0$.
3. **Modeling Implication:** The hourly order series is stationary in its raw level form. Consequently, first-order differencing ($d=0$) was appropriate for ARIMA modeling.

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## 7. Chronological Train/Test Split

In time-series modeling, standard random cross-validation cannot be used because it destroys temporal ordering and causes **data leakage** (predicting the past using future data).

Instead, a strict **chronological split** was executed, reserving the final portion of time for evaluation.

| Dataset Split | Date Range | Hour Count | Percentage |
|---|---|---|---|
| **Training Set** | April 1, 2026 – August 25, 2026 | 3,513 hours | ~80% |
| **Test Set** | August 25, 2026 – September 30, 2026 | 879 hours | ~20% |
| **Total** | **April 1 – September 30, 2026** | **4,392 hours** | **100%** |

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## 8. ARIMA Methodology and Results

An **ARIMA($p,d,q$)** model combines three components:

- **AR ($p=1$):** AutoRegressive term using the immediate preceding hour's value ($Y_{t-1}$).
- **I ($d=0$):** Integrated term representing the order of differencing (0, since the series is already stationary).
- **MA ($q=1$):** Moving Average term using the lagged residual error ($e_{t-1}$).

### Model Specification & Diagnostic Output

- **Model Structure:** ARIMA(1, 0, 1) with Constant
- **Estimated Constant (Intercept):** 1.1249 (matches the training population mean of ~1.14)
- **AR(1) Coefficient $p$-value:** 0.930
- **MA(1) Coefficient $p$-value:** 0.895
- **Akaike Information Criterion (AIC):** 10509.148
- **Ljung–Box Test (Lag 1) $p$-value:** 0.99

### Technical Evaluation & Diagnostics Explanation

- **Coefficient Significance:** The high $p$-values for the AR(1) term (0.930) and MA(1) term (0.895) indicate that these specific lagged parameters are **not statistically significant** at standard levels ($\alpha = 0.05$). The model effectively collapses close to a baseline mean-forecaster with noise.
- **Residual Autocorrelation:** The Ljung–Box test at lag 1 yields $p = 0.99$, indicating no significant autocorrelation in immediate lag 1 residuals.
- **Cautionary Note:** Achieving a high Ljung-Box $p$-value at lag 1 does **not** mean all residual autocorrelation across higher seasonal lags (e.g., 24-hour or 168-hour cycles) is absent. Therefore, this baseline ARIMA(1,0,1) model should **not** be considered production-ready.

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## 9. Prophet Methodology and Results

**Prophet** is an open-source forecasting framework developed by Meta designed for business time series featuring strong seasonal effects and historical trends.

### Model Configuration

- **Seasonality Settings:** Explicitly enabled daily and weekly seasonal components.
- **Growth:** Linear growth trend.
- **Handling Low Counts:** Modeled continuous output, evaluated on the 879-hour test window.

### Verified Test Performance

- **MAE:** 0.866 orders/hour
- **RMSE:** 1.124 orders/hour
- **MAPE (Positive-Demand Hours):** 32.126%

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## 10. Seasonal-Naive Baseline

To ensure our machine learning models deliver actual business value, we benchmarked them against a **168-hour Seasonal Naive baseline**.

### Baseline Logic

The Seasonal Naive approach predicts that demand for any given hour will be exactly equal to the demand observed at the same hour of the same day in the previous week:
$$\hat{Y}_{t} = Y_{t-168}$$

### Verified Test Performance

- **MAE:** 1.231 orders/hour
- **RMSE:** 1.614 orders/hour
- **MAPE (Positive-Demand Hours):** 71.998%

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## 11. Model Comparison Table and Visual Interpretation

The table below summarizes the test set evaluation across all three models.

| Model Name | MAE (orders/hr) | RMSE (orders/hr) | MAPE (Positive Demand Hours) | Performance vs. Baseline |
|---|---|---|---|---|
| **ARIMA(1,0,1)** | **0.860** | **1.122** | **31.712%** | **30.1% MAE Reduction** |
| **Prophet** | 0.866 | 1.124 | 32.126% | 29.6% MAE Reduction |
| **Seasonal Naive (168h)** | 1.231 | 1.614 | 71.998% | *Benchmark* |

### Explanation of Metric Definitions

1. **MAE (Mean Absolute Error):** The average absolute difference between predicted and actual orders per hour. An MAE of 0.860 means our model is off by less than 1 order per hour on average.
2. **RMSE (Root Mean Squared Error):** The square root of squared errors. RMSE penalizes larger prediction errors more heavily than MAE. The closeness of ARIMA's RMSE (1.122) to its MAE (0.860) indicates few extreme outlier errors.
3. **MAPE (Mean Absolute Percentage Error):** The average percentage error.
    - *Special Handling:* Standard MAPE is mathematically undefined when actual demand $Y_t = 0$ (division by zero). Therefore, **MAPE was computed exclusively on positive-demand hours ($Y_t > 0$)**.

### Comparative Takeaways

- **ARIMA vs. Prophet:** ARIMA(1,0,1) achieved a marginally lower MAE (0.860 vs 0.866) and RMSE (1.122 vs 1.124) than Prophet, but the performance difference between them is negligible (< 0.01 orders/hour).
- **ML vs. Baseline:** Both predictive models significantly outperformed the Seasonal Naive approach, reducing forecast error by over **30%**.
- **Understanding the High MAPE (~31–32%):** In low-volume datasets ($\mu = 1.14$), small absolute errors yield large percentage errors. For instance, if actual demand is 1 order and the model predicts 2, the absolute error is just 1 order, but the percentage error is $100%$.

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## 12. Business Insights

Translating these experimental results into operational context yields several actionable insights:

1. **Low Baseline Volatility:** Average order volume across the overall platform footprint is sparse (~1.14 orders/hour). Centralized fleet dispatches should avoid over-allocating fixed driver shifts based on unadjusted global averages.
2. **Buffer Capacity Management:** With an MAE under **0.9 orders/hour**, operational planners can maintain a tight standby driver buffer (+1 driver per zone) to handle potential forecast deviations without incurring excess labor costs.
3. **Baseline Replacement:** Replacing traditional rules-of-thumb (like last week's same-hour demand) with ARIMA or Prophet cuts forecast variance by ~30%, directly reducing driver idle time during low-demand periods.

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## 13. Limitations and Risks

1. **Count Data Granularity:** Order demand consists of discrete, non-negative integers (0, 1, 2, ...). Standard ARIMA and Prophet treat target variables as continuous real numbers, which can produce fractional predictions (e.g., predicting 1.43 orders) or negative point estimates during extreme low-trough periods.
2. **Parameter Inefficiency in Baseline ARIMA:** The high $p$-values for $AR(1)$ and $MA(1)$ coefficients indicate that the single-lag ARIMA structure is inefficiently parameterized and requires seasonal extension (SARIMA).
3. **Lack of External Drivers (Exogenous Variables):** The current models rely purely on historical timestamp patterns. They do not incorporate real-time weather alerts, localized promotions, or public holiday flags, which heavily impact demand spikes.

4. **Single Train-Test Split:** Performance was evaluated on a single 879-hour test split. Performance could vary across different seasonal transitions (e.g., winter vs. summer).

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## 14. Conclusion

Day 5 of the capstone project successfully established a quantitative time-series forecasting framework for Uber Eats hourly order demand.

Starting from raw transaction logs, we resampled 4,392 continuous hourly slots, confirmed stationarity via the ADF test ($p < 0.001$), and evaluated multiple modeling approaches on an 879-hour holdout dataset.

Both ARIMA(1,0,1) and Prophet demonstrated strong performance, achieving an average prediction error of ~0.86 orders/hour and delivering a **>30% accuracy improvement** over the Seasonal Naive baseline. While parameter non-significance in ARIMA(1,0,1) indicates the need for seasonal extensions (SARIMA) and count-data algorithms, this analysis establishes a benchmark for data-driven operational decision-making.